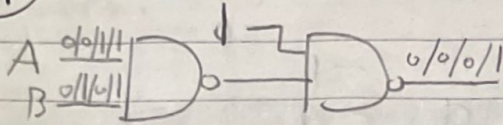
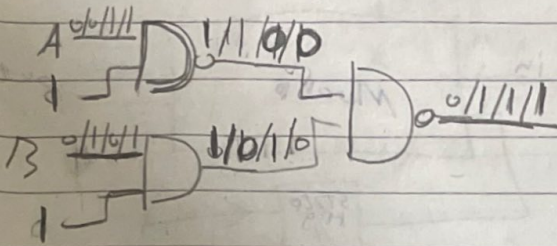


Logic design

① Build AND gate using NAND gate:

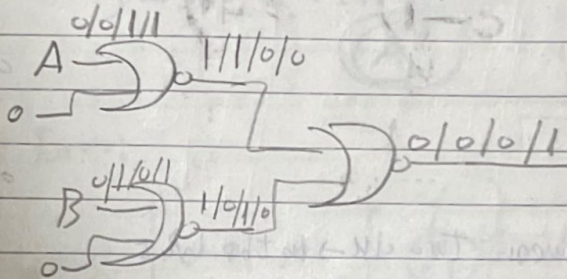


② Build OR gate using NAND gate:

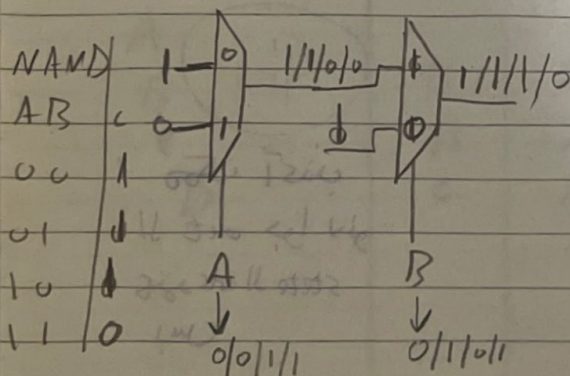


③ Build AND gate using NOR gate:

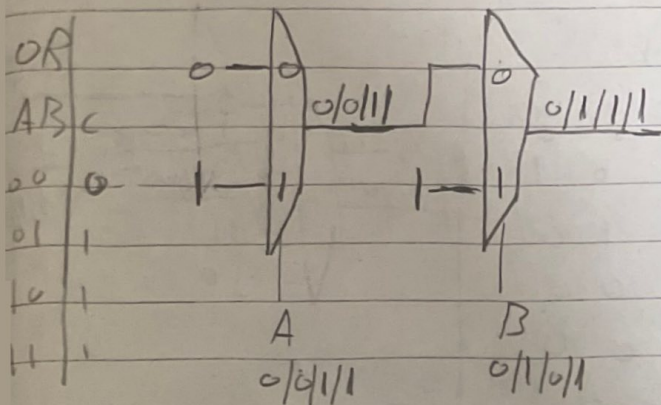
NOR	AND
A B C	A B C
0 0 1	0 0 0
0 1 0	0 1 0
1 0 0	1 0 0
1 1 0	1 1 1



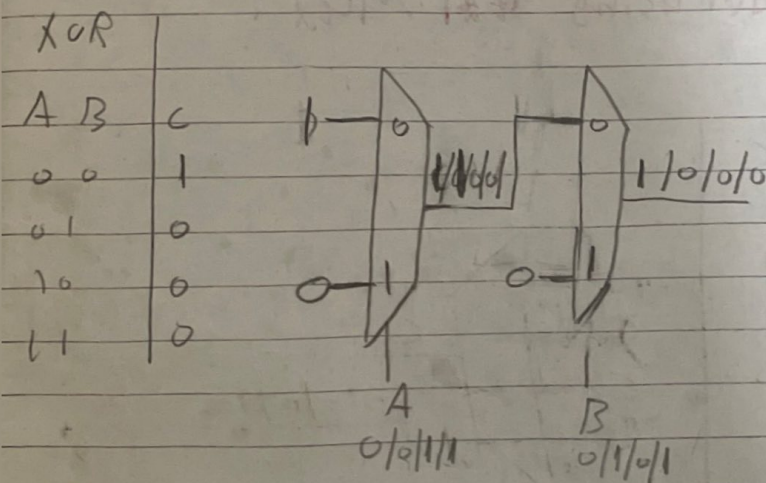
④ Build NAND gate using MUX:



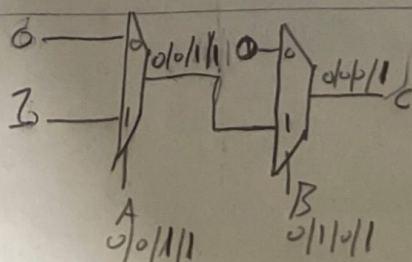
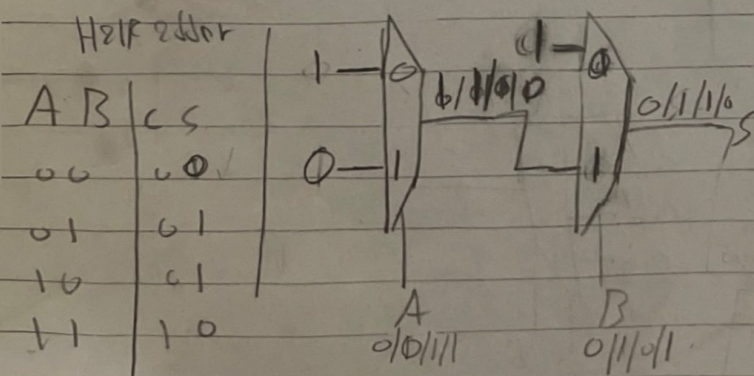
5 Build OR gate using Mux:



6 Build XOR gate using Mux:

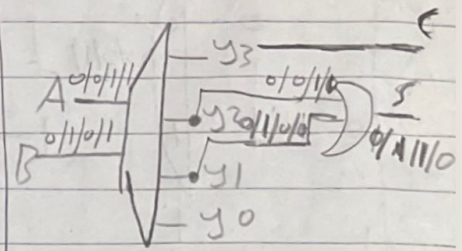


7 Build half adder using 2x1 Mux



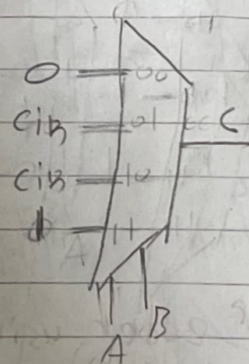
⑧ Build half adder using 2x4 Decoder:

half adder			2x4 decoder					
A	B	CS	A	B	y ₃	y ₂	y ₁	y ₀
0	0	00	0	0	0	0	0	1
0	1	01	0	1	0	0	1	0
1	0	01	1	0	0	1	0	0
1	1	10	1	1	1	0	0	0

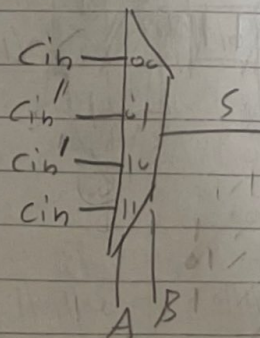


⑨ Build Full Adder using 4x1 Mux:

Full Adder				
A	B	Cin	Cout	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



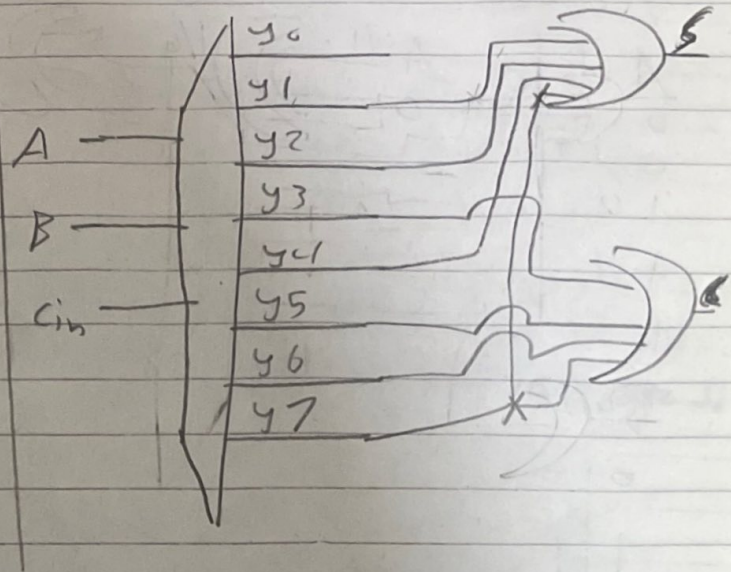
B	C	A
0	0	0 → 00 → 0
0	1	0 → 10 → 0
1	0	0 → 10 → 0
1	1	0 → 00 → 0
0	0	1 → 01 → 0
0	1	1 → 11
1	0	1 → 11
1	1	1 → 01



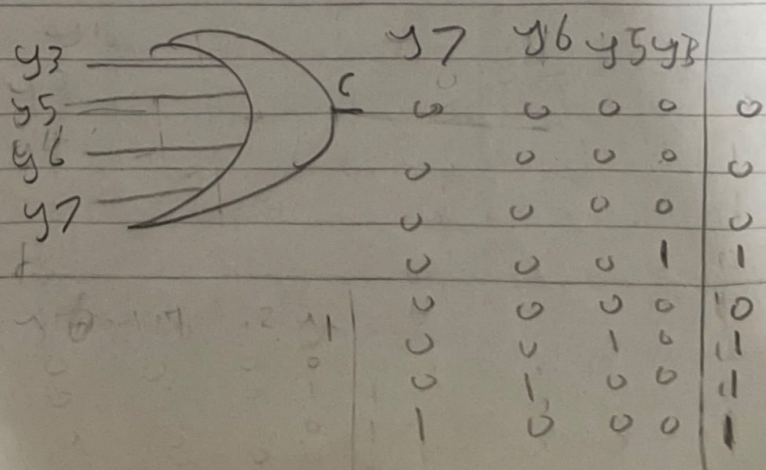
10 Build Full Adder using 3x2 Decoder:

Full adder

A	B	cin	cout	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



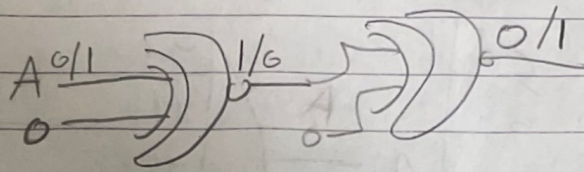
A	B	C	y7	y6	y5	y4	y3	y2	y1	y0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0



(11) How to use XNOR gate as a buffer?

XNOR

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1



Buffer

A	B
0	0
1	1